BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021 • Paper Code: 3471 / 3478

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q. Paper : I (Objective Type)

سوالیہ پرچہ: I (معروضی طرز)

QQ-1. 1021 — (جماعت دہم) (Session 2017-19 to 2019-21) — Inter Part-II — Sargodha Board — (معروضی) فزکس / Physics (Objective) — Group 1

Q. Time Allowed: 15 minutes — PAPER CODE 3471 — Maximum Marks: 12

Q1. If the mass of the bob of a pendulum is increased by a factor of 3, The period of the pendulum's motion will

(A) (B)

(C) (D)

Q2. Which is an example of a longitudinal wave?

(A) (B)

(C) (D)

Q3. Which type of image is formed by a Convex Lens on a screen?

(A) (B)

(C) (D)

Q4. The coulomb's law is valid for the charges which are

(A) (B)

(C) (D)

Q5. What is the voltage across a 6 Ω resistor when 3A of current passes through it

(A) (B)

(C) (D)

Q6. If the current in a wire placed perpendicular to magnetic field increases, the force on wire

(A) (B)

(C) (D)

Q7. The output of a NAND gate is 0 when ____

(A) (B)

(C) (D)

Q8. From which of the following we can get information almost about everything

(A) (B)

(C) (D)

Q9. Energy coming from the Sun and Stars is supposed to be the result of:

(A) (B)

(C) (D)

Q10. Refractive index of ice is.

(A) (B)

(C) (D)

Q11. What would happen to the loudness of sound with increase its frequency.

(A) (B)

(C) (D)

Q12. Electron volt is also a unit of energy used in atomic and Nuclear Physics which is equal to.

(A) (B)

(C) (D)

Board typos kept verbatim: 'Moving and or non-point charges' (Q4 option B), 'the coulomb's law'. Q1 option D, Q6 option D and Q12 option D are partly obscured by the scan watermark — unreadable fragments marked [illegible in source scan]. The objective is printed in a bilingual table (Urdu + English); questions transcribed from the printed English column.

Paper : I (Essay Type)

پرچہ: I (انشائیہ طرز)

QPart I. 1021 — (جماعت دہم) (Session 2017-19 to 2019-21) — Inter Part-II — Sargodha Board — (انشائیہ) فزکس / Physics (Essay Type) — Group 1

Q. Time Allowed: 1:45 hours — Maximum Marks: 48

QPart II. Part II — حصہ دوم

Note: Attempt any Two Questions. 9×2=18

Board typos kept verbatim: 'mutual Induction', 'back ground radiations', 'Simple Harmonic Motion' (mixed case), 'Time period equation', 'A point charge of +2C', 'Answer briefly any Five parts from the followings'. Subjective parts transcribed from the printed English column; the identical Urdu column appears in the scans as printed.

GROUP – II

Q. Paper : I (Objective Type)

سوالیہ پرچہ: I (معروضی طرز)

QQ-1. 1021 — (جماعت دہم) (Session 2017-19 to 2019-21) — Inter Part-II — Sargodha Board — (معروضی) فزکس / Physics (Objective) — Group 2

Q. Time Allowed: 15 minutes — PAPER CODE 3478 — Maximum Marks: 12

Q1. When a heavy nucleus splits into two lighter nuclei, the process would

(A) (B)

(C) (D)

Q2. Has less penetrating power.

(A) (B)

(C) (D)

Q3. The unit of capacitance is

(A) (B)

(C) (D)

Q4. If T = 1.99 sec then frequency is

(A) (B)

(C) (D)

Q5. In a vacuum, all electromagnetic waves have the same

(A) (B)

(C) (D)

Q6. The loudness of sound is more closely related to its

(A) (B)

(C) (D)

Q7. Which type of image is formed by a convex lens on a screen?

(A) (B)

(C) (D)

Q8. Electric field lines

(A) (B)

(C) (D)

Q9. What is the power rating of a lamp connected to a 12 V source when it carries 2.5 A?

(A) (B)

(C) (D)

Q10. The step up transformer

(A) (B)

(C) (D)

Q11. X = A.B then X = 1 when

(A) (B)

(C) (D)

Q12. From which of the following can get information about everything

(A) (B)

(C) (D)

Board typos kept verbatim: 'Has less penetrating power.' (Q2, no subject), 'The unit of capacitance is' (Q3), 'X = A.B then X = 1 when' (Q11), 'can get information about everything' (Q12), '0.50 Hz' printed for both options A and C in Q4. The objective is printed in a bilingual table (Urdu + English); questions transcribed from the printed English column.

Paper : I (Essay Type)

پرچہ: I (انشائیہ طرز)

QPart I. 1021 — (**جماعت دہم**) (Session 2017-19 to 2019-21) — Inter Part-II — Sargodha Board — (**انشائیہ**) **فزکس** / Physics (Essay Type) — Group 2

Q. Time Allowed: 1:45 hours — Maximum Marks: 48

QPart II. Part II — حصہ دوم

Note: Attempt any Two Questions. 9×2=18

Board typos kept verbatim: 'State lenz's law' (lowercase l), 'Write equation', 'Write the names of two dielectric', 'four web Browsers', '6v' (lowercase v), 'burglar alarm', 'Answer briefly any Five parts from the followings'. Subjective parts transcribed from the printed English column; the identical Urdu column appears in the scans as printed.